

these causal metrics, including attribution patching (Syed, Rager, and Conmy 2023) and improved versions thereof (Kramár et al. 2024; Hanna, Pezzelle, and Belinkov 2024; Marks et al. 2025). Even in the absence of a deep understanding of the role of these mediators, localization can be useful for downstream applications like model editing (Meng et al. 2022, 2023)²⁹ and model steering (Todd et al. 2024; Goyal et al. 2020). That said, if meaningful features are not actually aligned with neurons/heads, then we are not guaranteed to get the best performance unless we use more fine-grained and selective mediators. Future work should analyze the performance of model editing and steering methods when using different kinds of mediators. For example, Marks et al. (2025) compare the efficacy of debiasing approaches based on ablating neurons versus non-basis-aligned directions discovered via sparse autoencoders; they find that ablating non-basis-aligned directions is significantly more effective. Wu et al. (2025), however, find that supervised approaches for locating non-basis-aligned directions (e.g., difference-in-means, Marks and Tegmark 2023) are significantly more effective than sparse autoencoders. These studies represent promising initial steps toward the kind of principled comparison between mediators that we advocate.

7.2 Suggestions for Future Work

By centering the mediator type and the criteria defined in §2.1, we can gain new insights into the kinds of research that will be necessary to advance mechanistic interpretability. Here, we discuss lines of work we believe will be fruitful.

7.2.1 Finding better causal mediators. There are almost certainly better causal mediators that have not yet been explored. By “better”, we mean achieving a better Pareto optimum between the criteria described in §2.1 for at least one of the listed goals. Current work on improving mediators tends to focus on non-basis-aligned directions, such as sparse features or directions discovered from supervised probing on the activations of a single layer/submodule h^l . One could consider pursuing coarser-grained mediators by discovering multi-layer **model regions** or **component sets** that accomplish a single behavior. Because these regions can cross layers, they would include non-linearities that allow them to represent more complex functions or concepts. We believe this would improve the generality of our findings, but not necessarily sparsity nor human-interpretability. Thus, we believe coarser-grained mediators will be better-suited to verifying mechanistic hypotheses or localization and editing.

Non-linear and multidimensional feature discovery. As discussed in §5.4, there is recent work demonstrating the existence of human-interpretable *multidimensional* features. For example, days of the week are encoded circularly as a set of 7 directions in a two-dimensional subspace (Engels et al. 2025), and current methods cannot easily capture these multidimensional features. The ability to discover these features could greatly improve our understanding of the feature space of NNs, and thus our ability to systematically explain more of their behavior (i.e., increase generality) in a faithful and human-interpretable way. Group sparse autoencoders (Theodosios and Ba 2023) or clusters of autoencoder features could be a way to capture multidimensional non-basis-aligned features in an unsupervised manner, but despite promising initial evidence, empirical

²⁹ Though Hase et al. (2023) find that causal localizations do not always reflect the optimal locations for editing models.

work has not yet demonstrated to what degree this will be effective for interpreting or controlling NNs. Additionally, many current mechanistic interpretability methods require us to define binary distinctions between correct and incorrect answers, whereas causal mediation analysis does not have any theoretical linearity, dimensionality, or Boolean restrictions; thus, interpretability methods will need to be extended to handle new kinds of variables.

There may also exist higher-order non-linear concepts in latent space that we have not yet located due to the linear focus of contemporary methods. For example, a subgraph or subcircuit can encode a coherent variable representation or functional role, as in Lepori, Serre, and Pavlick (2024); Li, Davies, and Nadeau (2024). How can we discover these subgraphs? Path patching (Goldowsky-Dill et al. 2023; Wang et al. 2023) provides a manual approach to implicating subgraphs as causal mediators, but we do not yet have automatic methods that can scalably search over subgraphs of a computation graph. Even if we were able to locate them, how might non-linear and/or coarse-grained mediators like these be useful in practice? As an example, we might expect fundamental phenomena like syntax processing to be spread across many layers of a language model. Syntax-sensitive components should be implicated in downstream tasks like question answering (QA) if we expect that language models are robustly parsing the meaning of the inputs. Thus, one could use causal mediation analysis to locate all components in the model with some high indirect effect on syntactic processing (e.g., using a subject–verb agreement task); this can be conceptualized as the syntax processing region of the model. Then, one could implicate the region(s) in the model’s QA performance by intervening on each component in the region and observing whether performance changes. If the syntax processing region is not strongly implicated in QA performance, then we have a strong hint that the model may not be parsing the meaning of the inputs, but instead relying on a mixture of surface-level spurious heuristics—for example, memorized bigram associations, or giving answers with high prior probabilities. These examples illustrate how these more coarse-grained mediators could help us verify new kinds of mechanistic hypotheses.

7.2.2 Inherently interpretable models. More ambitiously, one could consider building models with inherently interpretable components—i.e., whose fundamental units of computation (or some subset thereof) are designed to be sparse, monosemantic, and/or human-interpretable, but ideally still expressive enough to attain good performance on downstream tasks. Examples based in neural networks include differentiable masks (De Cao et al. 2020; Bastings, Aziz, and Titov 2019), transcoders (Dunefsky, Chlenski, and Nanda 2024), codebook features (Tamkin, Taufeque, and Goodman 2023), and softmax linear units (Elhage et al. 2022a). These are primarily post hoc methods that decompose model components into interpretable units, but they could potentially be integrated into the network itself during pre-training alongside a loss term (in addition to a typical cross-entropy loss) that enables fine-grained interpretability at all stages of pre-training.

Alternatively, more focus could be devoted to building models that are designed from the ground up to be interpretable, such as backpack language models (Hewitt et al. 2023), concept bottleneck models (Koh et al. 2020; Oikarinen et al. 2023), or decision trees (Hu, Rudin, and Seltzer 2019). A related idea is to train the model using loss terms that encourage success on intermediate tasks, or induce particular kinds of feature representations (Hupkes, Zuidema et al. 2017). Perhaps least invasively, we could consider pre-training methods that softly encourage interpretable features to be aligned to neuron bases; this would remove the need for optimization to find non-basis-

aligned components, and therefore make interpreting model decisions significantly easier and less confounded. However, this would reduce the number of features that could be encoded per neuron, so it would likely require significantly more parameters, or accepting degradations in performance. Regardless, we believe that this line of work will improve our ability to explain the behaviors of contemporary NN-based systems like neural language models via directly improving the human-understandability of their intermediate computations; this will make it far easier to explain model behaviors, verify mechanistic hypotheses, and localize/edit particular computations—but at the potential expense of performance on downstream tasks.

7.2.3 Scalable search. As the size of neural networks increases, the number of potential mediators to search over will also increase. The situation worsens as we start searching over continuous sets of fine-grained mediators such as non-basis-aligned directions. Although a few gradient-based or optimization-based approximations to causal influence have been proposed to improve time efficiency, such as attribution patching (Syed, Rager, and Conmy 2023) and DCM (Davies et al. 2023), more work is still needed to evaluate the efficacy of these techniques in identifying the correct causal mediators. Additionally, better techniques beyond greedy search methods should be devised to identify causally important *groups of* mediators; these should aim to produce Pareto improvements over time complexity and causal efficacy.

As discussed in §6.2.1, optimization-based mediator search methods can be effective, but often require pre-existing hypotheses as to how a model implements a particular behavior. Thus, one could investigate using large language model agents to automate the process of hypothesis generation. Qiu et al. (2024) showed that current LLMs can generate hypotheses, and Shaham et al. (2024) showed that hypothesis refinement via LLMs can aid humans in interpreting the causal role of neurons in multimodal models. Similarly, LLMs could be used to automate and scale hypothesis generation regarding the role of particular mediators across a wider variety of tasks and models. Optimization-based methods such as DAS or DCM could then be used to causally verify the automatically generated hypotheses, potentially in an iterative loop of hypothesis refinement and empirical testing.

7.2.4 Benchmarking progress in mechanistic interpretability. How will we know when we have made genuine improvements along any of the criteria that we have proposed? There exist few standardized methods or datasets for measuring general progress. To address this, research is needed on **standard benchmarks for measuring progress in mechanistic interpretability**. Currently, most studies develop ad hoc evaluations, and only compare to similar methods that employ the same mediators. Thus, to measure whether new mediators or search methods are truly giving us improvements over previous ones, we need to develop methods for performing principled direct comparisons. In circuit discovery, it is theoretically possible to use the same metrics to compare any circuit discovered for a particular model and task, regardless of whether sparse autoencoders are used, whether the circuit is based on nodes or edges, among other variations. Some recent work has begun to perform direct comparisons across mediator types, such as Miller, Chughtai, and Saunders (2024). Huang et al. (2024) propose to directly evaluate interpretability methods according to the generality of the abstractions they recover, and directly compare across different mediator types given the same model and task. Arora, Jurafsky, and Potts (2024) and Makelov, Lange, and Nanda (2024) also propose standardized interpretability benchmarks that allow us to compare across mediator search methods, though they do not directly compare across mediator types.